Model Development Document

PFE / CCR Exposure Engine

Version 1.0

Apex Global Bank | Quantitative Research Division

SR 11-7 Section Reference: Model Development | March 2026

Classification: RESTRICTED — Model Risk Sensitive

1. Document Metadata

Field	Value
Model Name	PFE / CCR Exposure Engine
Model ID	APEX-MDL-0014
Version	1.0
Status	In Validation
Risk Tier	Tier 1
Regulatory Framework	Basel III SA-CCR; IFRS 13 Fair Value; SR 11-7
Date of Development	March 2026
Business Owner	Head of Global Markets Trading (James Okafor)
Model Developer	Quantitative Research — XVA Team
MRO Review	Pending — Dr. Rebecca Chen
CRO Approval	Pending — Dr. Priya Nair

Table 1: Document metadata

1.1 Version History

Version	Date	Change Summary
0.1	Jan 2026	Initial netting framework; EPE/ENE profiles only
0.8	Feb 2026	Collateral offset with MPoR; SA-CCR add-on module
1.0	Mar 2026	Production candidate; submitted for validation

Table 2: Version history

2. Executive Summary

2.1 Model Purpose

The PFE / CCR Exposure Engine is the foundational simulation model for all counterparty credit risk (CCR) measurement at Apex Global Bank. It generates the exposure profiles — Expected Positive Exposure (EPE), Expected Negative Exposure (ENE), and Potential Future Exposure (PFE) — that are consumed by three downstream Tier 1 models:

- **CVA** (APEX-MDL-0010): uses EPE to compute the market value of counterparty default risk;
- **FVA** (APEX-MDL-0011): uses EPE/ENE to compute the funding cost of uncollateralised positions;
- **MVA** (APEX-MDL-0012): uses simulated future Initial Margin profiles derived from the same Monte Carlo paths.

In addition, the model produces:

- **PFE for credit limit management**: the 97.5th-percentile exposure used by the Credit Risk function to set and monitor counterparty credit limits;
- **SA-CCR capital**: the Basel III standardised-approach CCR exposure-at-default (EAD) used for regulatory capital calculation.

Because this model is upstream of multiple Tier 1 models and directly feeds regulatory capital reporting, it is classified as **Tier 1**.

2.2 Output Description

Output	Consumer	Frequency
EPE profile (time series)	CVA, FVA, MVA models	Daily
ENE profile (time series)	FVA model	Daily
PFE at 97.5% (time series)	Credit limit system	Daily
Effective EPE (Basel)	Regulatory capital (IMM)	Daily
SA-CCR EAD by netting set	Capital reporting, RWA	Daily
Netting benefit report	Credit / legal	Monthly

Table 3: Model outputs and consumers

2.3 Key Assumptions

1. Market factors evolve under the risk-neutral measure (for XVA) and the real-world measure (for PFE / credit limits); the current implementation uses risk-neutral for all outputs with a real-world scaling factor applied to PFE percentiles.
2. Market factor dynamics follow the parametric models specified in Section **??** (Hull-White for rates, GBM for FX/equity, CIR for credit spreads).
3. Netting sets are legally isolated: exposure netting is applied within but not across ISDA Master Agreements.
4. The collateral posting party exercises its collateral obligations in full at each margin call date; no dispute or operational delay is modelled outside the Margin Period of Risk.
5. SA-CCR calculation follows BCBS 279 (March 2014) as implemented in US rules effective April 2020.

2.4 Known Limitations

1. The Gaussian copula used for joint factor simulation underestimates tail dependence in stress scenarios.
2. The correlation matrix is static and recalibrated annually; intra-year correlation shifts are not captured.

3. Gap risk for equity derivatives (overnight jumps) is not explicitly modelled.
4. Approximately 8% of counterparties require proxy CDS spread mapping, introducing basis risk.
5. Path-dependent exotic derivatives use computational approximations that may introduce pricing errors of up to 5%.

3. Business Context and Purpose

3.1 The Counterparty Credit Risk Problem

OTC derivatives expose Apex to the risk that a counterparty defaults when the derivative has positive mark-to-market value to Apex. Unlike a loan, where exposure is the outstanding principal, a derivative's exposure is a random variable that depends on the future evolution of market prices. The fundamental challenge of CCR is that this exposure must be estimated over the entire life of a contract — potentially 30 years for a long-dated interest rate swap.

The 2008 financial crisis demonstrated the catastrophic consequences of underestimating CCR. Bear Stearns and Lehman Brothers had large derivatives portfolios with hundreds of counterparties. Exposure estimates that failed to account for netting, or that used inadequate correlation assumptions, produced dangerously low credit limits. When stress arrived, actual losses exceeded modelled estimates by factors of 3–5x.

3.2 Regulatory History

CCR modelling has been shaped by successive waves of Basel regulation:

- **Basel II (2004)**: introduced the Internal Model Method (IMM) allowing banks with approved simulation models to use EPE for CCR capital, replacing the cruder Current Exposure Method (CEM).
- **Basel III (2010)**: introduced the CVA capital charge requiring EPE-based CVA calculation; introduced the stressed EPE concept.
- **BCBS 279 / SA-CCR (2014)**: replaced CEM with the more risk-sensitive Standardised Approach to CCR, mandatory from 2020 for banks without IMM approval.
- **FRTB-CVA (2017)**: reformed the CVA capital framework, requiring banks to compute CVA sensitivities using an approved simulation infrastructure.

Apex currently uses SA-CCR for regulatory capital and maintains this model as a candidate for IMM approval, which would allow use of internal EPE for capital purposes.

3.3 Model Users

4. Theoretical Foundation

4.1 Close-Out Netting

Under an ISDA Master Agreement, all transactions between two parties within a single netting set are aggregated upon default. The close-out netting value at time t is:

$$V_{\text{net}}(t) = \sum_{k=1}^{K} V_k(t) \tag{1}$$

User	Function	Output Used
Credit Risk Officers	Counterparty limit setting and monitoring	PFE at 97.5%
XVA Desk	CVA/FVA/MVA pricing and hedging	EPE, ENE
Regulatory Capital	Basel III SA-CCR RWA calculation	SA-CCR EAD
Finance / P&L	CVA P&L via downstream CVA model	EPE
Internal Audit / MRO	Model validation and ongoing monitoring	All outputs
Regulators (OCC, Fed)	CCAR stress testing; SR 11-7 review	SA-CCR EAD, EPE

Table 4: Model users and primary outputs consumed

where $V_k(t)$ is the mark-to-market value of transaction k at time t, measured from Apex's perspective (positive = Apex is owed money).

The *netting benefit* at time t quantifies how much the netting agreement reduces gross positive exposure:

$$\text{NettingBenefit}(t) = \sum_{k=1}^{K} \max(V_k(t),\, 0) - \max\left(\sum_{k=1}^{K} V_k(t),\, 0\right) \tag{2}$$

A netting benefit of zero implies either all trades are in the same direction (no offset) or the netting set is entirely out-of-the-money for Apex.

4.2 Collateral Offset and the CSA

Where a Credit Support Annex (CSA) governs variation margin, the posted collateral reduces the effective exposure at each simulation date. Let $C(t)$ denote the collateral balance (positive = collateral held by Apex), Θ the CSA threshold, and δ the minimum transfer amount (MTA). The collateralised exposure before MPoR adjustment is:

$$V_{\text{col}}(t) = \max\big(V_{\text{net}}(t) - C(t) - \Theta - \delta,\; 0\big) \tag{3}$$

For a fully collateralised CSA with $\Theta = 0$ and $\delta = 0$, the exposure collapses to $\max(V_{\text{net}}(t) - C(t), 0)$.

4.3 Margin Period of Risk

The Margin Period of Risk (MPoR) is the time from the last successful margin call to the close-out of the portfolio following a counterparty default. During MPoR, Apex cannot call additional margin; the last received collateral may be stale. BCBS 261 specifies:

- **10 business days**: daily-margined netting sets (standard);
- **20 business days**: netting sets where the counterparty has $\geq$ 5,000 transactions or illiquid collateral;
- **At least 20 business days**: re-hypothecated collateral.

The effective collateral at simulation time t, accounting for MPoR delay τ, is:

$$C_{\text{eff}}(t) = C(t - \tau) \tag{4}$$

where τ is the applicable MPoR. Substituting into (**??**):

$$V_{\text{col,MPoR}}(t) = \max\big(V_{\text{net}}(t) - C(t-\tau) - \Theta - \delta,\ 0\big) \tag{5}$$

4.4 Exposure Metrics

Let $\tilde{V}(t) \equiv V_{\text{col,MPoR}}(t)$ denote the collateralised, MPoR-adjusted exposure. All exposure metrics are derived from the distribution of $\tilde{V}(t)$ across Monte Carlo paths.

4.4.1 Expected Exposure and Expected Positive / Negative Exposure

$$\text{EE}(t) = \mathbb{E}\Big[\max\big(\tilde{V}(t),\ 0\big)\Big] \tag{6}$$

$$\text{EPE} = \frac{1}{T}\int_0^T \text{EE}(t)\,dt \tag{7}$$

$$\text{ENE}(t) = \mathbb{E}\Big[\min\big(\tilde{V}(t),\ 0\big)\Big] \tag{8}$$

The EPE in (**??**) represents the time-averaged expected exposure and is the primary input to the CVA calculation.

4.4.2 Effective EPE (Regulatory)

For Basel IMM purposes, the *Effective Expected Exposure* is the non-decreasing envelope of EE, ensuring the regulatory capital measure is conservative with respect to roll-off of near-term exposure:

$$\text{EffEE}(t) = \max\big(\text{EE}(t),\ \text{EffEE}(t-\Delta t)\big), \quad \text{EffEE}(0) = V_{\text{net}}(0) \tag{9}$$

The *Effective EPE* is then:

$$\text{EffEPE} = \frac{1}{1\text{Y}}\int_0^{1\text{Y}} \text{EffEE}(t)\,dt \tag{10}$$

4.4.3 Potential Future Exposure

PFE at confidence level α at time horizon t is the α-quantile of the exposure distribution:

$$\text{PFE}_\alpha(t) = Q_\alpha\Big[\tilde{V}(t)\Big] \tag{11}$$

Apex uses $\alpha = 97.5\%$ as the standard PFE confidence level for credit limit purposes, consistent with Basel counterparty credit risk guidance.

4.5 Market Factor Models

4.5.1 Interest Rates: Two-Factor Hull-White

Domestic and foreign short rates are modelled using the two-factor Hull-White (HW2) model, which fits the initial term structure exactly while allowing mean-reverting stochastic evolution:

$$r(t) = x_1(t) + x_2(t) + \varphi(t) \tag{12}$$

$$dx_1(t) = -a_1\, x_1(t)\, dt + \sigma_1\, dW_1(t) \tag{13}$$

$$dx_2(t) = -a_2\, x_2(t)\, dt + \sigma_2\, dW_2(t) \tag{14}$$

where $\varphi(t)$ is the deterministic fit to the initial forward curve, a_1, a_2 are mean-reversion speeds, σ_1, σ_2 are volatilities, and $\mathrm{Corr}(dW_1, dW_2) = \rho_{12}$.

Current calibration (March 2026): $a_1 = 0.03$, $a_2 = 0.35$, $\sigma_1 = 0.007$, $\sigma_2 = 0.014$, $\rho_{12} = -0.72$.

4.5.2 FX Rates: Geometric Brownian Motion

FX spot rates $S_{\mathrm{CCY/USD}}(t)$ are modelled as:

$$\frac{dS(t)}{S(t)} = \big(r_d(t) - r_f(t)\big)\, dt + \sigma_{\mathrm{FX}}\, dW_{\mathrm{FX}}(t) \tag{15}$$

where r_d and r_f are domestic and foreign short rates from the HW2 models, and σ_{FX} is the FX volatility calibrated to ATM options with maturity matching the simulation horizon.

4.5.3 Equity: Heston Stochastic Volatility

Equity prices are modelled under the Heston (1993) framework to capture the volatility smile:

$$\frac{dS(t)}{S(t)} = \mu\, dt + \sqrt{v(t)}\, dW_S(t) \tag{16}$$

$$dv(t) = \kappa\big(\bar{v} - v(t)\big)\, dt + \xi\sqrt{v(t)}\, dW_v(t) \tag{17}$$

with $\mathrm{Corr}(dW_S, dW_v) = \rho_{\mathrm{SV}}$. Parameters are calibrated to the equity volatility surface.

4.5.4 Credit Spreads: Cox-Ingersoll-Ross

Credit spreads (used for CVA hazard rate evolution) follow a CIR process, ensuring non-negativity:

$$d\lambda(t) = \kappa_\lambda\big(\bar{\lambda} - \lambda(t)\big)\, dt + \sigma_\lambda\sqrt{\lambda(t)}\, dW_\lambda(t) \tag{18}$$

4.6 Joint Simulation via Cholesky Decomposition

All market factors are simulated jointly using a 47×47 correlation matrix $\mathbf{P}$, Cholesky-decomposed into $\mathbf{L}$ such that $\mathbf{L}\mathbf{L}^\top = \mathbf{P}$. Standard normal variates $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ are transformed into correlated variates $\mathbf{w} = \mathbf{L}\mathbf{z}$, which are then used to drive the stochastic differential equations above.

The 47 factors comprise: 5 interest rate curves (SOFR, ESTR, SONIA, TONAR, HIBOR), 10 FX pairs (major and EM currencies), 12 equity indices, 15 credit spread curves (investment grade, high yield, financials, sovereigns), and 5 commodity indices.

4.7 SA-CCR: Standardised Approach to Counterparty Credit Risk

For regulatory capital purposes under Basel III (BCBS 279), the Exposure-at-Default (EAD) for a netting set is:

$$\mathrm{EAD} = \alpha \cdot \big(\mathrm{RC} + \mathrm{PFE}_{\mathrm{addon}}\big) \tag{19}$$

where $\alpha = 1.4$ is the supervisory scalar, RC is the replacement cost, and $\mathrm{PFE}_{\mathrm{addon}}$ is the supervisory add-on.

4.7.1 Replacement Cost

For margined netting sets:

$$\mathrm{RC} = \max\big(V - C,\ \mathrm{TH} + \mathrm{MTA} - \mathrm{NICA},\ 0\big) \tag{20}$$

For unmargined netting sets:

$$\mathrm{RC} = \max\big(V - C,\ 0\big) \tag{21}$$

where V is current net MtM, C is collateral held net of haircuts, TH is the CSA threshold, MTA is the minimum transfer amount, and NICA is the net independent collateral amount.

4.7.2 PFE Add-On Aggregation

The aggregate PFE add-on across all five asset classes (interest rate, FX, credit, equity, commodity) is:

$$\mathrm{PFE}_{\mathrm{addon}} = \mathrm{AddOn}^{\mathrm{IR}} + \mathrm{AddOn}^{\mathrm{FX}} + \mathrm{AddOn}^{\mathrm{credit}} + \mathrm{AddOn}^{\mathrm{EQ}} + \mathrm{AddOn}^{\mathrm{com}} \tag{22}$$

Each asset-class add-on is computed from the supervisory delta, notional, maturity factor, and supervisory volatility as specified in BCBS 279 Annex B. For interest rate derivatives, the add-on within a single currency hedging set uses the partial netting formula:

$$\mathrm{AddOn}^{\mathrm{IR,\ CCY}} = \left| \sum_j \delta_j \cdot d_j \cdot \mathrm{MF}_j \cdot \sigma_j^{\mathrm{IR}} \right| \tag{23}$$

where $\delta_j \in \{+1, -1\}$ is the supervisory delta, d_j is the adjusted notional, MF_j is the maturity factor ($\mathrm{MF}_j = \sqrt{\min(M_j, 1\mathrm{Y})/1\mathrm{Y}}$ for unmargined), and σ_j^{IR} is the supervisory volatility (= 0.50% for IR).

5. Data Requirements and Governance

5.1 Input Data Sources

Data Item	Source	Frequency
OIS/RFR yield curves (SOFR, ESTR, SONIA, TONAR, HIBOR)	Bloomberg / Internal	Daily
FX spot rates (10 pairs vs. USD)	Bloomberg	Real-time (E
Equity index levels (12 indices)	Bloomberg	Daily
Credit spread curves (IG, HY, fins, sov)	Markit / Bloomberg	Daily
Swaption vol surfaces (all rate curves)	Bloomberg	Daily
FX ATM vol (all pairs)	Bloomberg	Daily
Equity vol surfaces (all indices)	Bloomberg	Daily
Credit spread vol (CDS options where available)	Markit	Weekly
ISDA Master Agreement database	Legal / ISDA operations	On amendme
CSA terms (threshold, MTA, eligible collateral)	Legal / ISDA operations	On amendme
Trade population (all OTC derivatives)	Front-office systems	Real-time
Collateral balances (VM posted/received)	Collateral Management	Daily
Historical factor data (5-year lookback)	Bloomberg / Proprietary	Annual

5.2 Data Quality Controls

1. **Curve completeness**: All OIS curves must have at least 10 tenor points covering 1W to 30Y. Missing tenors use linear log-discount interpolation with a staleness flag if the bracketing points are > 2 business days old.
2. **Vol surface consistency**: Swaption vol surfaces are validated for absence of calendar spread arbitrage and butterfly spread arbitrage before use. Surfaces failing validation revert to prior-day with an alert.
3. **Trade data completeness**: Trade population must reconcile to front-office systems within 0.5% of notional before the simulation run proceeds. Material breaks ($> 1\%$) halt the run and escalate to Operations.
4. **CSA database**: The CSA terms database must cover $\geq 99\%$ of live netting sets by notional. Missing CSA terms default to "uncollateralised" (conservative).

5.3 BCBS 239 Compliance

As a Tier 1 model feeding regulatory capital, the PFE/CCR engine is subject to BCBS 239 data lineage requirements. Every input data item is tagged with provenance metadata (source system, timestamp, transformation applied). The full data lineage from raw market data to SA-CCR EAD must be reconstructable within 4 hours of a regulatory request.

6. Methodology and Implementation

6.1 Monte Carlo Engine

The simulation engine generates $N = 10{,}000$ risk-neutral paths over $M = 200$ time steps:

- $t_1, \ldots, t_{52}$: weekly steps to 1 year;
- $t_{53}, \ldots, t_{112}$: monthly steps from 1 to 5 years;

- $t_{113}, \ldots, t_{200}$: quarterly steps from 5 to 30 years.

GPU acceleration (CUDA via cuBLAS) is used for the 47×47 Cholesky decomposition and for batched path generation. The full portfolio run completes in under 4 hours on the production GPU cluster. An intraday "Greeks" run (500 paths, 1Y horizon only) completes in under 30 minutes.

6.2 Netting Set Aggregation Algorithm

For each simulation path ω and time step t:

1. Price all trades $V_k(\omega, t)$ using the simulated market factors.
2. Group trades by ISDA netting set identifier.
3. Compute $V_{\text{net}}(\omega, t) = \sum_k V_k(\omega, t)$ within each netting set (Equation **??**).
4. Apply collateral offset per Equation **??**, using the CSA terms for the netting set.
5. Record $\tilde{V}(\omega, t)$ as the path-level collateralised exposure.

6.3 Exposure Profile Construction

After all N paths are generated, for each netting set and time step t:

1. $\text{EE}(t) = \frac{1}{N} \sum_\omega \max(\tilde{V}(\omega, t), 0)$;
2. $\text{ENE}(t) = \frac{1}{N} \sum_\omega \min(\tilde{V}(\omega, t), 0)$;
3. $\text{PFE}_{97.5\%}(t) = 97.5\text{th-percentile of } \{\tilde{V}(\omega, t)\}_{\omega=1}^{N}$;
4. $\text{EffEE}(t)$ computed recursively per Equation **??**.

Portfolio-level profiles aggregate netting-set profiles using the same path-consistent summation (no diversification benefit across netting sets for EPE, consistent with Basel IMM requirements).

6.4 SA-CCR Calculation

The SA-CCR module runs as a post-processing step on the current-date trade population (no simulation paths required). For each netting set:

1. Classify each trade into asset class (IR, FX, credit, EQ, commodity).
2. Compute supervisory delta δ_j for each trade (sign and optionality adjustment).
3. Compute adjusted notional d_j (trade-specific; tenor-adjusted for IR).
4. Compute maturity factor MF_j (margined or unmargined formula).
5. Aggregate add-ons within hedging sets (with partial netting for IR/credit).
6. Aggregate hedging-set add-ons to netting-set $\text{PFE}_{\text{addon}}$.
7. Compute RC per Equation **??** or **??**.
8. Compute EAD per Equation **??**.

6.5 Shared Infrastructure with CVA, FVA, MVA

CVA, FVA, and MVA all consume outputs from the same Monte Carlo run. This ensures consistency across XVA measures — a requirement under IFRS 13 for coherent fair value measurement — and reduces computational cost by avoiding redundant simulation. The dependency chain is:

$$\textbf{PFE/CCR Engine} \longrightarrow \textbf{CVA}\ (\text{APEX-MDL-0010})$$
$$\textbf{PFE/CCR Engine} \longrightarrow \textbf{FVA}\ (\text{APEX-MDL-0011})$$
$$\textbf{PFE/CCR Engine} \longrightarrow \textbf{MVA}\ (\text{APEX-MDL-0012})$$

A failure of the PFE/CCR engine propagates to all three downstream models. The fallback procedure (prior-day EPE/ENE scaled by Greeks) is documented in Section **??**.

7. Model Testing and Backtesting

7.1 P&L Explain for EE

Daily predicted EE change is compared to actual EE change:

$$\Delta\text{EE}_{\text{predicted}} = \frac{\partial \text{EE}}{\partial r}\Delta r + \frac{\partial \text{EE}}{\partial S_{\text{FX}}}\Delta S_{\text{FX}} + \frac{\partial \text{EE}}{\partial \lambda}\Delta\lambda + \text{theta} \tag{24}$$

P&L explain ratio $= \Delta\text{EE}_{\text{predicted}}/\Delta\text{EE}_{\text{actual}}$ must exceed 85%. Deviations $> 15\%$ trigger a model investigation within 2 business days.

7.2 SA-CCR Benchmark

The SA-CCR EAD is compared quarterly to the 97.5th-percentile internal PFE multiplied by $\alpha = 1.4$ (the SA-CCR scalar):

$$\text{SA-CCR ratio} = \frac{\text{SA-CCR EAD}}{1.4 \times \text{PFE}_{97.5\%}(1\text{Y})} \tag{25}$$

Acceptable range: [0.9, 1.5]. Values outside this range indicate either over-conservatism in SA-CCR or inadequacy of the internal model, and trigger a 30-day review.

7.3 Historical Backtesting

EE predictions are compared to realised exposures on a rolling 1-year lookback. The test asks: "did the realised 1-year maximum exposure fall below the predicted $\text{PFE}_{97.5\%}$ at least 97.5% of the time?" Following the Basel traffic light framework:

Zone	Exceptions (out of 250 obs.)	Action
Green	0–6	No action required
Yellow	7–9	Explanation required; potential add-on
Red	10+	Model review mandatory; capital add-on

Table 6: Backtesting traffic light zones (adapted for CCR)

7.4 Stress Scenario Analysis

Five stress scenarios are applied to the current portfolio and the resulting EE profiles are reported to the Market Risk Officer quarterly:

Scenario	EE Increase ($M)	$PFE_{97.5\%}$ Increase ($M)
2008 GFC replay	+$1,100	+$2,400
2020 COVID stress	+$650	+$1,500
Rates +200bp parallel shift	+$420	+$980
Credit spreads +100bp	+$280	+$620
FX +/-20% (dollar weakening)	+$210	+$450

Table 7: Stress scenario EE and PFE impacts (approximate, current portfolio)

8. Performance Metrics and Calibration Standards

Metric	Acceptable Range	Frequency	Action if Breached
EE P&L Explain	$> 85\%$	Daily	Investigate within 2 business day
SA-CCR ratio (SA-CCR / 1.4×PFE)	[0.9, 1.5]	Quarterly	Review within 30 days
Backtesting exceptions (annual)	≤ 6 (Green zone)	Quarterly	Escalate if Yellow; mandatory if
Correlation matrix condition #	$< 1{,}000$	Annual	Recalibrate; notify MRO
Data completeness (CSA coverage)	$\geq 99\%$	Daily	Escalate to Legal/Operations
Intraday run completion time	< 30 minutes	Daily	Technology escalation
Full EOD run completion time	< 4 hours	Daily	Technology escalation

8.1 Recalibration Schedule

- **HW2 and CIR parameters**: Annual recalibration to 5-year historical data; ad hoc recalibration upon material market structure change (e.g., rate regime change).
- **Heston equity parameters**: Quarterly recalibration to current vol surface.
- **Correlation matrix**: Annual recalibration using rolling 5-year equally-weighted data; reviewed after market stress events.
- **SA-CCR supervisory parameters**: Fixed by BCBS 279; updated only upon regulatory change with full MRO review.

9. Limitations, Assumptions, and Known Weaknesses

9.1 Gaussian Copula Tail Dependence

The joint factor simulation uses a Gaussian copula: correlated normal random variables are mapped to each factor's marginal distribution. The Gaussian copula underestimates tail dependence — the probability of multiple factors simultaneously moving to extreme values is lower in the Gaussian copula than in empirical data, particularly during systemic stress events.

Compensating control: Stressed EE (computed using 2008 and 2020 shock scenarios) is disclosed alongside base EE. A capital add-on for tail dependence is included in the model reserve.

9.2 Static Correlation Matrix

The 47×47 correlation matrix is calibrated once per year. During periods of market stress, correlations can change rapidly and materially (e.g., equity-FX correlations invert during risk-

off events). The static matrix may produce inaccurate exposure estimates intra-year.

Compensating control: The Market Risk Officer monitors intraday correlation estimates from a 60-day rolling window and can trigger an unscheduled recalibration if any major pair moves > 30 percentage points.

9.3 Equity Gap Risk

The Heston model uses continuous-time diffusion with no jump component. Overnight equity gaps (e.g., a 20% single-day drop following an unexpected corporate event) are not captured. For equity derivatives with short-dated optionality, this can understate PFE.

Compensating control: Equity derivatives with remaining maturity < 6 months use a jump-adjusted vol ($\sigma_{\text{adj}} = 1.2 \times \sigma_{\text{Heston}}$) for PFE calculation only.

9.4 Proxy Credit Spread Mapping

Approximately 8% of counterparties by notional have no observable CDS spread. These use a proxy mapping based on credit rating and sector (e.g., BBB-rated European utility $\mapsto$ iTraxx Main component average). The proxy spread may not reflect idiosyncratic credit risk.

Compensating control: Proxy-mapped counterparties are flagged in all outputs. A 25% add-on is applied to the EPE for proxy-mapped counterparties in the CVA calculation. The Credit Risk function reviews proxy assignments quarterly.

9.5 Exotic Derivative Approximations

Path-dependent exotic derivatives (e.g., barrier options, Asian options, TARFs) are priced using closed-form approximations or pre-computed exposure grids rather than full nested Monte Carlo. These approximations may introduce pricing errors of up to 5% for individual instruments.

Compensating control: Exotics exceeding \$50M in notional receive a full nested Monte Carlo re-price on a monthly basis. The largest 20 exotic positions are benchmarked to an independent pricing library quarterly.

10. Compensating Controls

Risk	Control	Owner
Tail dependence underestimation	Stressed EE disclosure; model reserve add-on	
Static correlation	60-day rolling monitor; unscheduled recalibration trigger at 30pp move	
Equity gap risk	1.2× vol adjustment for short-dated equity derivatives	
Proxy CDS mapping	25% EPE add-on; quarterly review by Credit Risk	
Exotic approximation error	Monthly full reprice for >\$50M; quarterly benchmark vs. independent	
Engine failure (downstream cascade)	Fallback: prior-day EPE/ENE scaled by Greeks; alert to CVA/FVA/M	
CSA database gaps	Missing CSA defaults to uncollateralised (conservative); nightly reconc	
Data staleness	2-business-day staleness alert; fallback to prior-day with flag	

10.1 Model Reserve

A model reserve of **\$35 million** is held against PFE/CCR model uncertainty:

- Gaussian copula tail dependence: $15M;
- Proxy CDS mapping basis risk: $10M;
- Exotic derivative path-dependency approximation: $10M.

The reserve is reviewed annually alongside the model recalibration cycle and may be adjusted following material market events.

11. Relationship to Downstream Models

The PFE/CCR Exposure Engine is a producer model; it does not consume outputs from other models within the XVA suite. The table below summarises the downstream dependencies:

Downstream Model	Model ID	Input Consumed	Sensitivity
CVA	APEX-MDL-0010	EPE profile (all counterparties)	High — linear in EPE
FVA	APEX-MDL-0011	EPE and ENE profiles	High — linear in EPE/E
MVA	APEX-MDL-0012	Simulated IM profiles (Monte Carlo)	High — direct path depe
ColVA	APEX-MDL-0013	|EPE| for collateral balance	Medium

Table 10: Downstream model dependencies

Any change to the PFE/CCR engine that affects EPE or ENE profiles (e.g., correlation matrix recalibration, factor model change, netting set re-aggregation) triggers a mandatory impact assessment for all four downstream models before the change is deployed to production.

12. Change Management

Material changes requiring full MRO re-validation:

- Change to any market factor model (HW2 → LMM, Heston → SABR, etc.);
- Change to netting set aggregation logic or MPoR treatment;
- Change to SA-CCR add-on calculation methodology;
- Integration of new asset class or new market (e.g., adding crypto derivative exposure);
- Migration to a new simulation infrastructure or hardware platform.

Non-material changes via expedited review (5 business days):

- Annual recalibration of factor model parameters (within pre-approved bounds);
- Addition of a new counterparty to the proxy CDS mapping table;
- Computational optimisation with no change to outputs;
- Bug fixes that correct errors without changing methodology.

13. Approvals and Sign-Off

Document Classification: RESTRICTED — Model Risk Sensitive
Apex Global Bank | Quantitative Research Division | XVA Team
Next scheduled review: March 2027 (or upon material model change)

Role	Name	Status	Date
Model Developer	XVA Quant Team	✓ Submitted	March 2026
Model Risk Officer	Dr. Rebecca Chen	○ Pending	—
Chief Risk Officer	Dr. Priya Nair	○ Pending	—
Business Owner	James Okafor	○ Pending	—

Table 11: Approval status

A. Notation Summary

Symbol	Definition
$V_k(t)$	MtM value of trade k at time t (Apex's perspective)
$V_{\text{net}}(t)$	Net MtM value of netting set at time t
$C(t)$	Collateral balance at time t (positive = held by Apex)
Θ	CSA threshold
δ	CSA minimum transfer amount (MTA)
τ	Margin Period of Risk
$\tilde{V}(t)$	Collateralised, MPoR-adjusted exposure
$\text{EE}(t)$	Expected Exposure at time t
EPE	(Time-averaged) Expected Positive Exposure
$\text{EffEE}(t)$	Effective Expected Exposure (non-decreasing)
$\text{ENE}(t)$	Expected Negative Exposure at time t
$\text{PFE}_\alpha(t)$	Potential Future Exposure at confidence α, horizon t
α	SA-CCR supervisory scalar ($= 1.4$)
RC	SA-CCR Replacement Cost
EAD	Exposure at Default (SA-CCR)
N	Number of Monte Carlo paths ($= 10{,}000$)
M	Number of simulation time steps ($= 200$)
a_1, a_2	Hull-White mean-reversion speeds
σ_1, σ_2	Hull-White volatilities
$\kappa, \bar{v}, \xi$	Heston mean-reversion, long-run variance, vol-of-vol
$\kappa_\lambda, \bar{\lambda}, \sigma_\lambda$	CIR credit spread parameters
$\mathbf{P}$	47×47 factor correlation matrix
$\mathbf{L}$	Cholesky factor of $\mathbf{P}$